

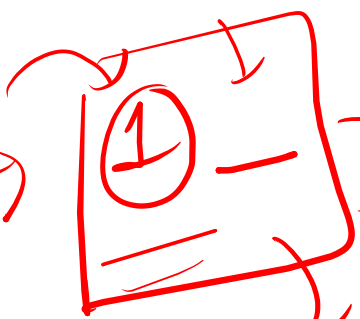
$\left\{ \begin{array}{l} \text{print}("111"); \\ \downarrow \\ \times 3 \end{array} \right\}$

$\left. \begin{array}{l} \rightarrow 1 \ 1 \ 1 \\ \rightarrow 1 \ 1 \ 1 \\ \rightarrow 1 \ 1 \ 1 \end{array} \right\} 100$

simplest way.

nested loop //

100 times

(i) $\rightarrow$  $\rightarrow$ 100 times automated?

$\downarrow \downarrow \swarrow$
 $\text{for}(i=1; i \leq 100; i++) \{$

$\quad \text{for}(j=1; j \leq 100;$

$\quad \quad j++)$
 $\quad \quad \text{print}("1");$
 $\quad \}$

$$n=4$$

$$i=1 \rightarrow 1$$

$$i=2 \rightarrow 1\ 2$$

$$i=3 \rightarrow 1\ 2\ 3$$

$$i=4 \rightarrow 1\ 2\ 3\ 4$$

$$i \rightarrow \begin{array}{c} 1 \dots \textcircled{i} \\ \underline{1\ 2\ 3} \dots \textcircled{i} \end{array}$$

code

$$\hookrightarrow n=4$$

input

$$\rightarrow \begin{array}{c} 1 \\ 1\ 2 \\ 1\ 2\ 3 \\ 1\ 2\ 3\ 4 \end{array}$$

$$n=3$$

$$\rightarrow \begin{bmatrix} 1 \\ 1\ 2 \\ 1\ 2\ 3 \end{bmatrix}$$

line numbers → for(ⁿi=1; i<=n; i++) { n → changes

for(j=1; j<=i; j++) {

system.out.print(j + " ");

}

⊗ system.out.println(); ←

}

content

i=2: 2 // i-1

$n=3$ ✓

1

2 1

2 1

$i \rightarrow n-i$ spaces
 $i=1 \rightarrow$ reverse order //

$n=4$

✓	✓	✓	1
		2	1
	3	2	1
4	3	2	1

$i=1 \rightarrow 3$ spaces $\rightarrow 1$

$i=2 \rightarrow _ _ 2 1$

$i=3 \rightarrow _ 3 2 1$

$i=4 \rightarrow 4 3 2 1$

$i \rightarrow n-i$ spaces
 $i \rightarrow 1 \dots i$
 $i-1 \dots 1$

$n=4$

1			1				
2		1	2	1			
3	1	2	3	2	1		
4	1	2	3	4	3	2	1

$3 \rightarrow$
 1 space ✓
 1 2 3 $\rightarrow$ 1 $\dots$ i
 2 1 $\rightarrow$ i-1 $\dots$ 1

1 $\rightarrow$ 3 spaces 1 ✓
 2 $\rightarrow$ 2 spaces
 1 2
1 i
 1 i-1 $\dots$ 1

$n = 4$
 $i = 3 \rightarrow$

①

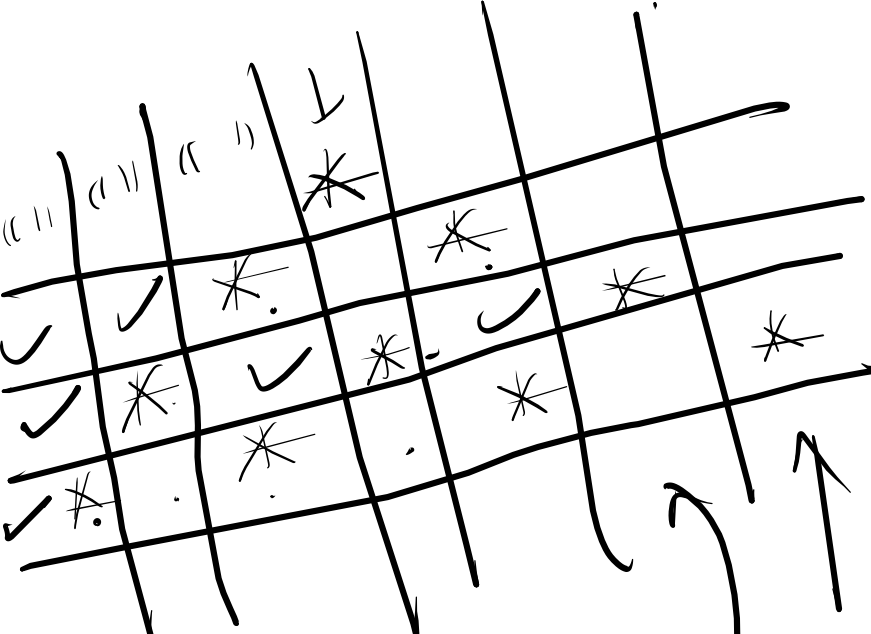
1
0 1)

1 2 3

2
2 1

1 ... i

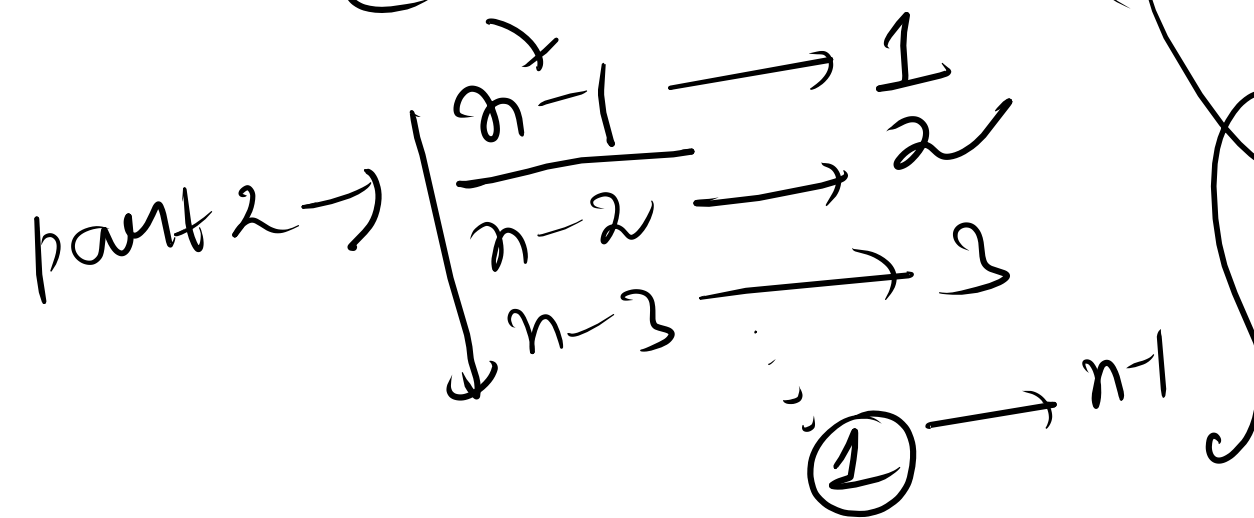
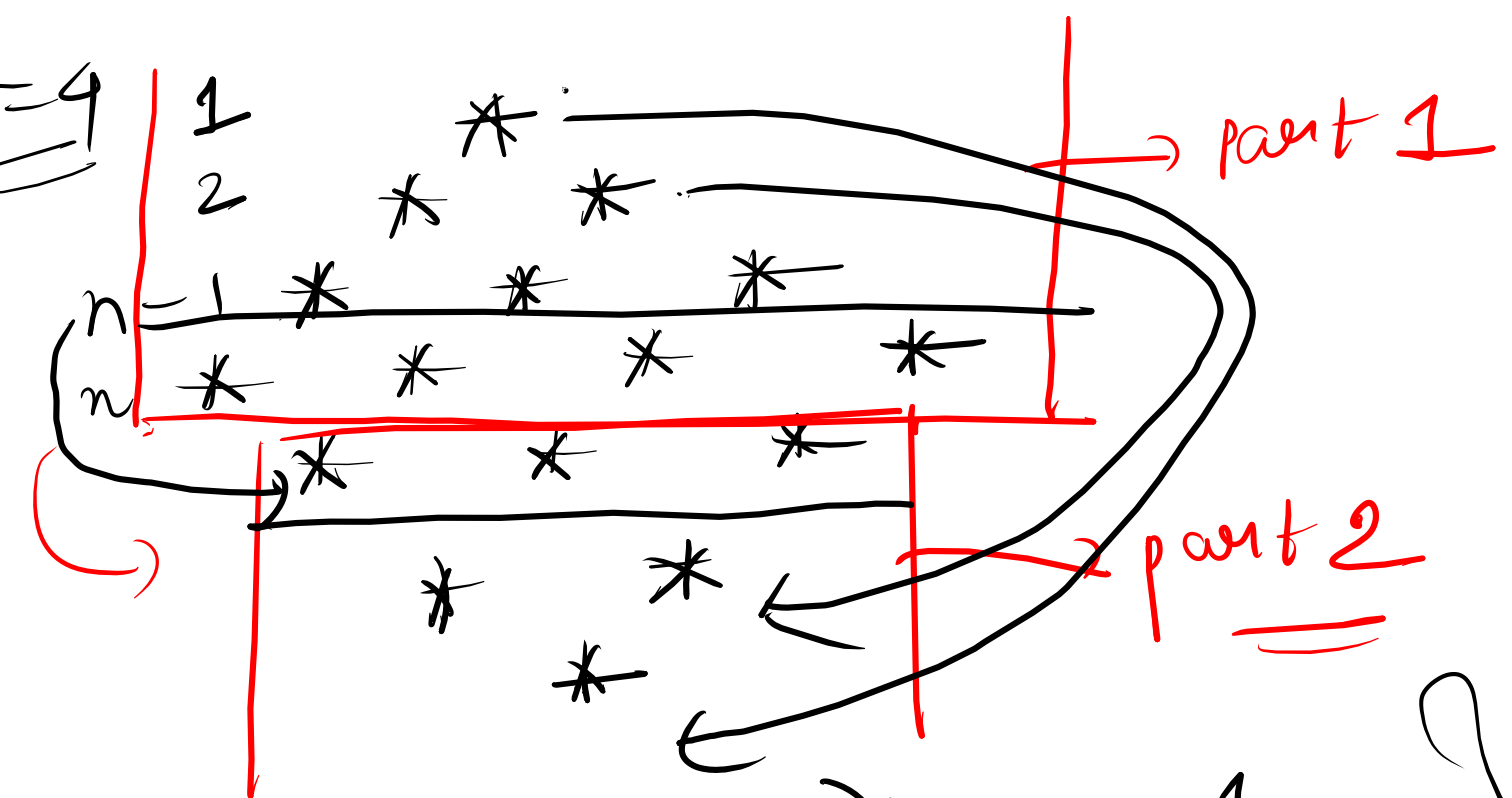
i-1 ... 1



$i = n - i$ spaces
 i stars



$n=4$



$n \times n$
 chess board

$n=4$

W	B	W	B
B	W	B	W
W	B	W	B
B	W	B	W

$n=3$

	1	2	3
1	W	B	W
2	B	W	B
3	W	B	W

$(r+c) \% 2 == 0$
 even
 ↓
 odd
 ↓
 B

5 * 4

num = 1

~~n =~~ { n = 5 → rows
m = 4
↓
cols. }

① 2 3 4
5 6 7 8
9 0 1 2
3 4 5 6
7 8 9 0

1 2 3 4
5 6 7 8
9 0 1 2
3

5 * 4

num / 10

⑪ → 1
⑫ → 2
⑬ → 3

n x n

2 * 4 * 4

i/s

i = 1/n

i/s
i = 1/n
j = 1/n

	1			n
1	1	1	1	1
	1	0	0	1
	1	0	0	1
n	1	1	1	1

→ print

```
for(i=1; i<=n; i++) {
```

```
    for(j=1; j<=n; j++) {
```

```
        else {
```

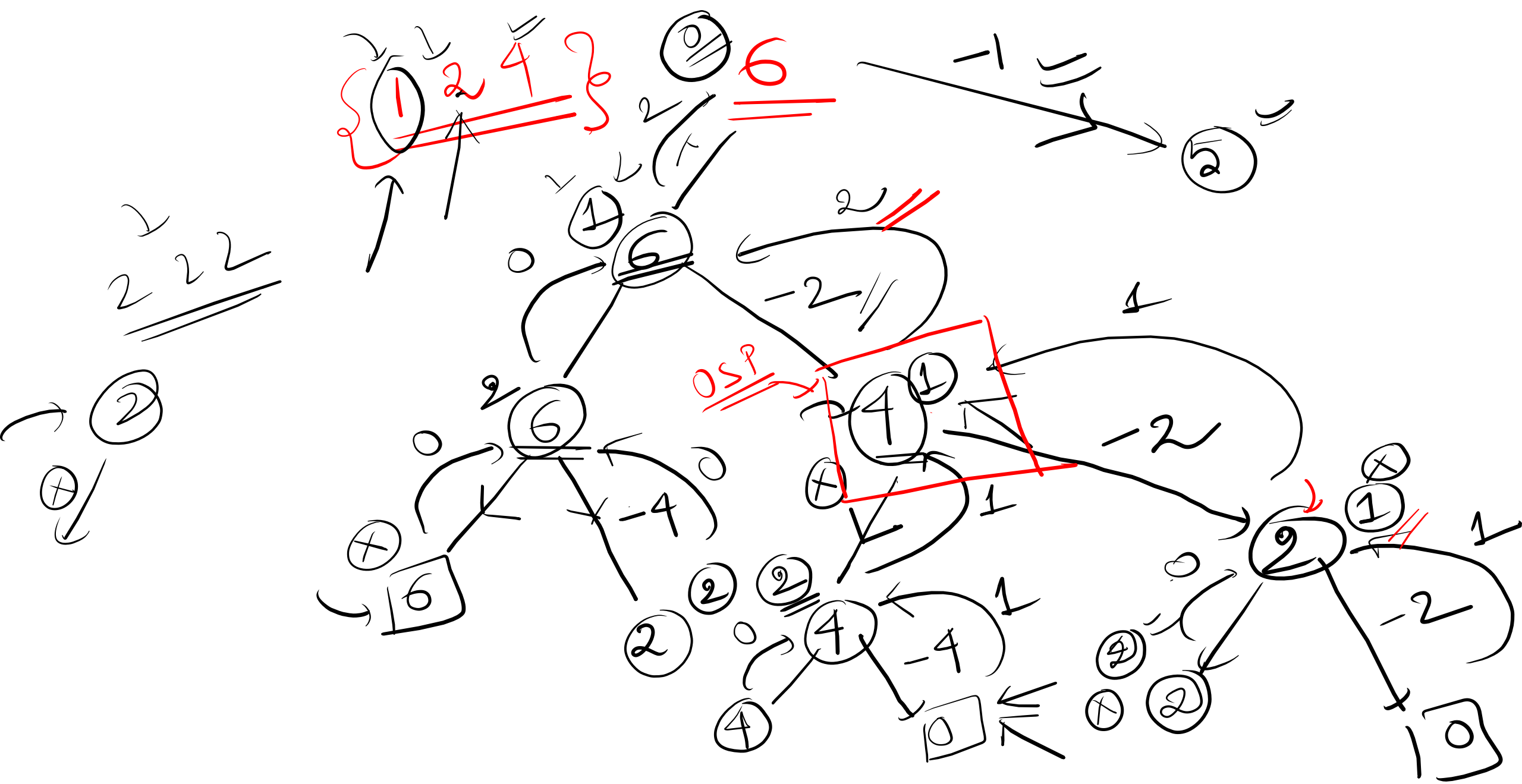
```
            print('1');
```

```
        }  
        println();  
    }
```

```
        if(i==1 || i==n ||
```

```
           j==1 || j==n)
```

```
            print('0');
```



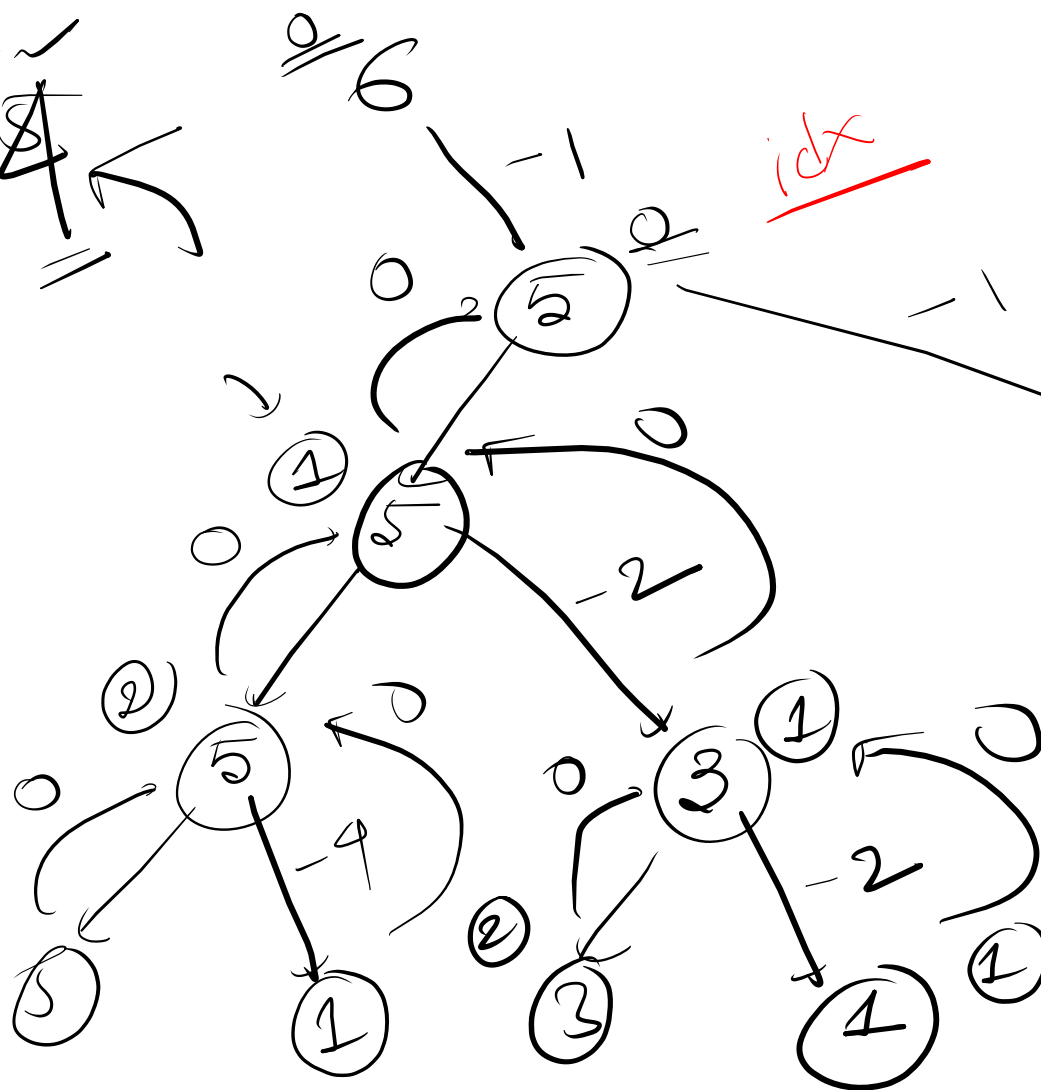
target $\rightarrow$ 16

1 2 3 4

0 1 2
1 2 3
4 5 6

unique ways

$f \rightarrow \frac{\text{sum}}{\text{target}}$ $\rightarrow \text{idx}$
 $\{0, 1, 2\}$

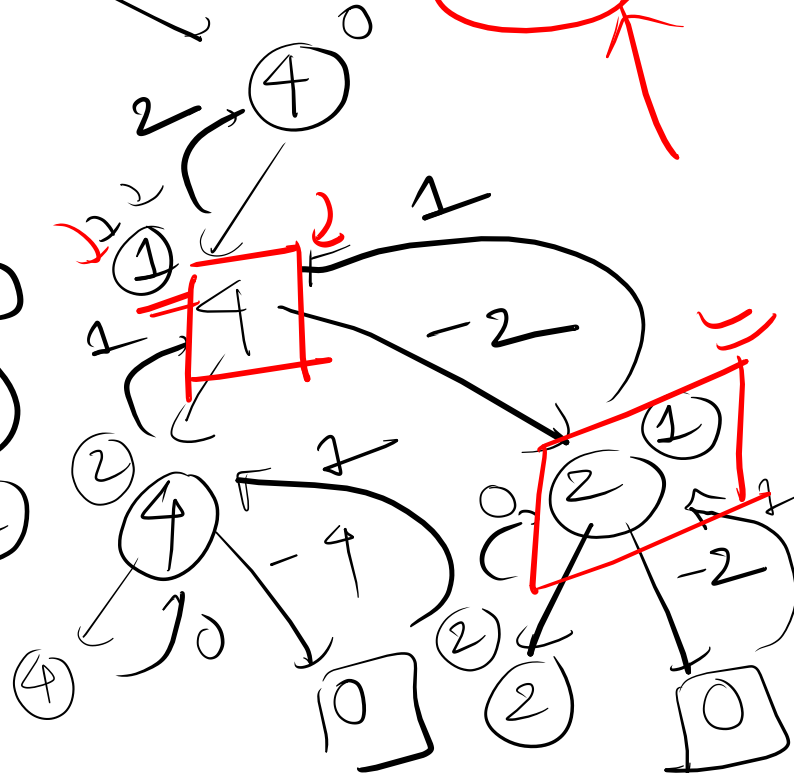


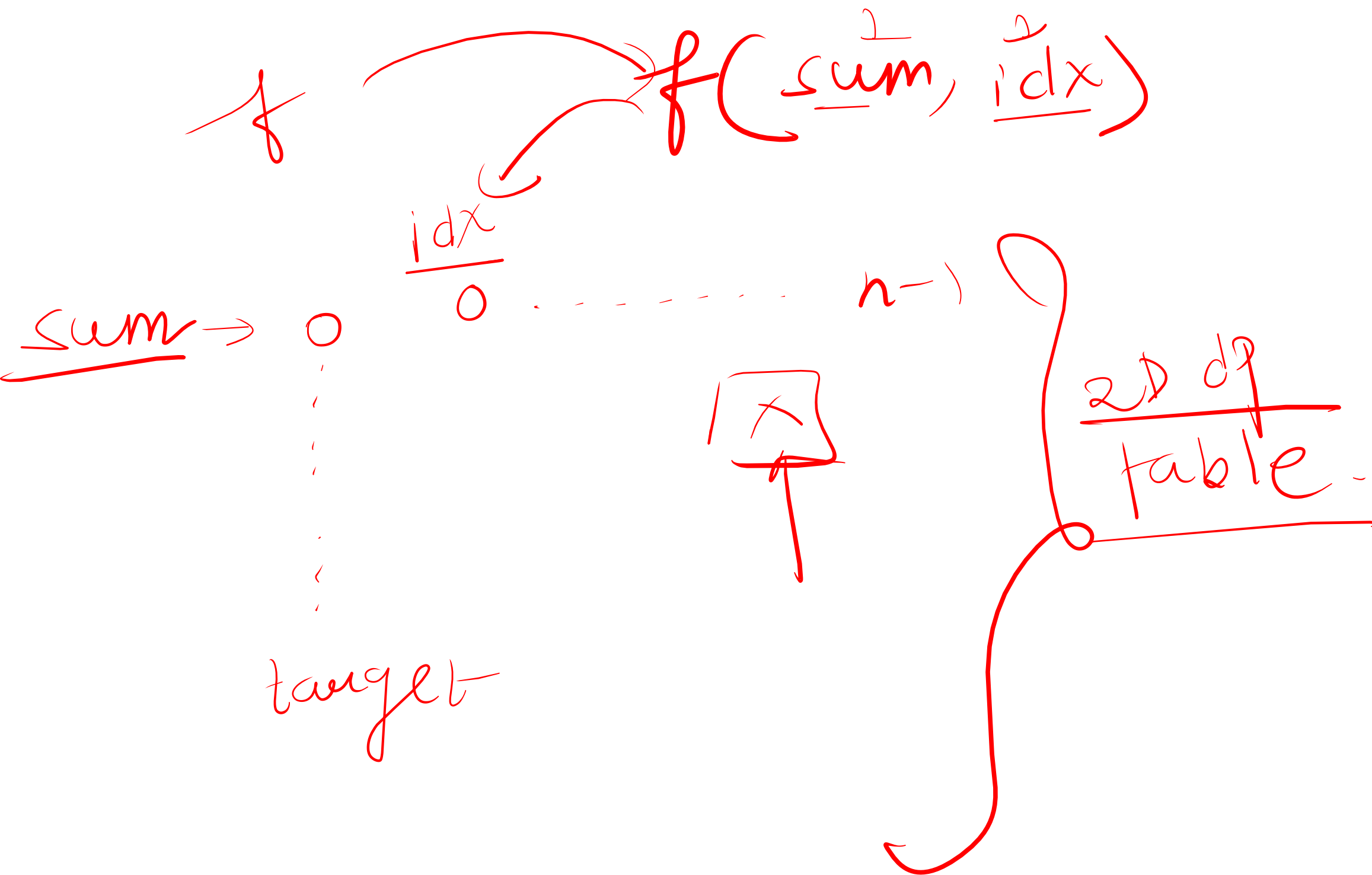
idx

sum

$f \rightarrow \text{idx, sum}$

sum





$\uparrow$
 $dp[sum+1][n];$
 $\downarrow$
 $\textcircled{-1}$

$w_1 = f(sum, idx+1);$

$w_2 = 0;$

if (coins[idx] == sum)

$w_2 = f(sum - coin[idx],$
 $idx);$

return $dp[sum][idx] = w_1 + w_2;$

}

$f(sum, idx) \{$

if (idx == n) {

if (sum == 0)

return 1;

return 0;

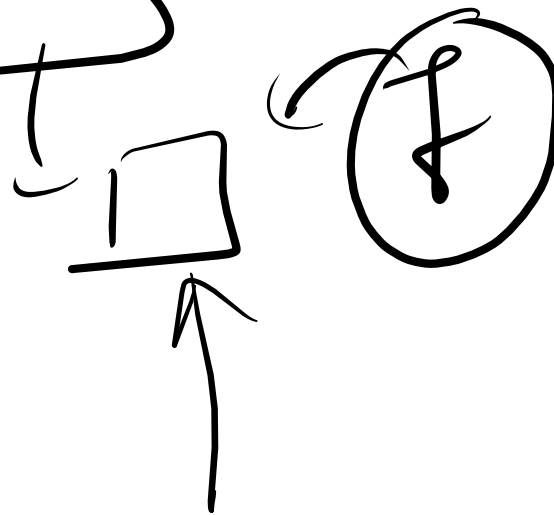
if ($\checkmark dp[sum][idx] \checkmark \neq -1$)

return $dp[sum][idx];$



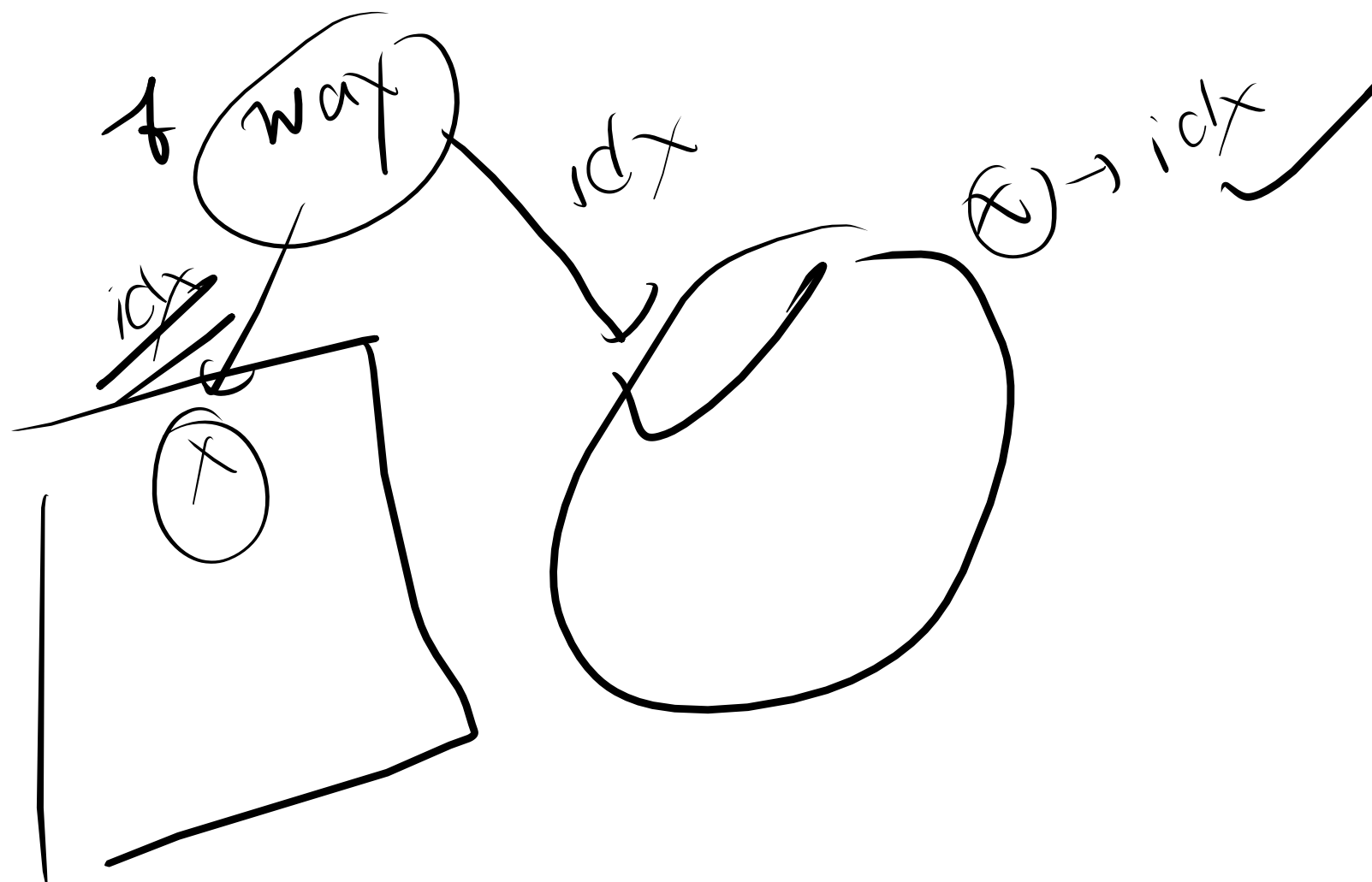
$O(\text{sum} * n)$
+ { recursive
overhead }
(X)

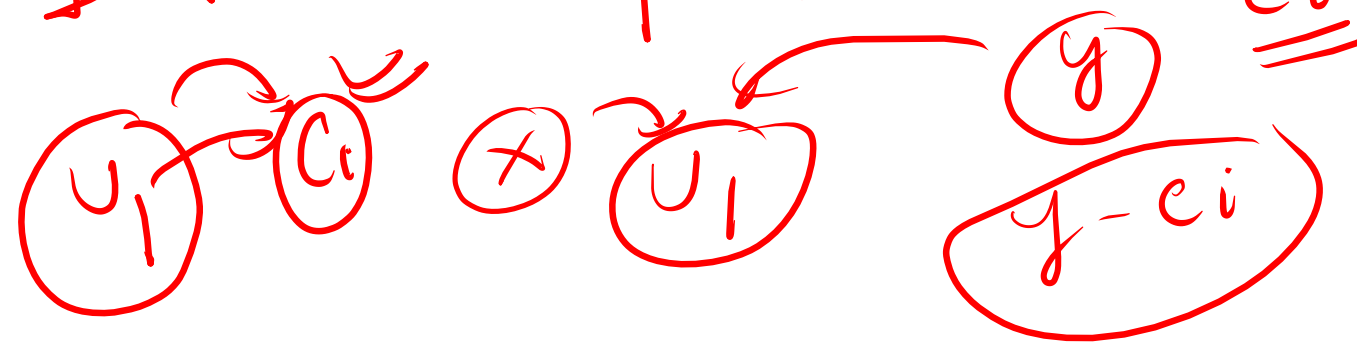
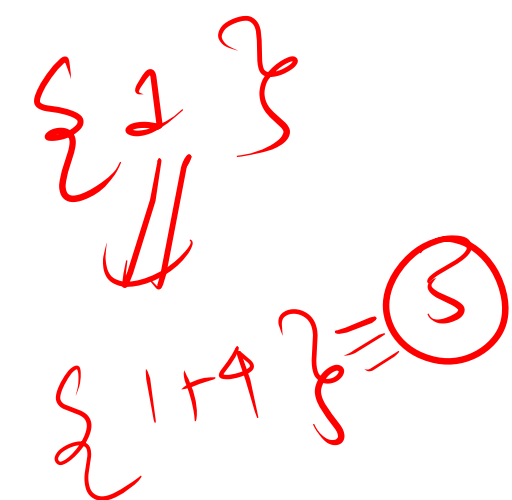
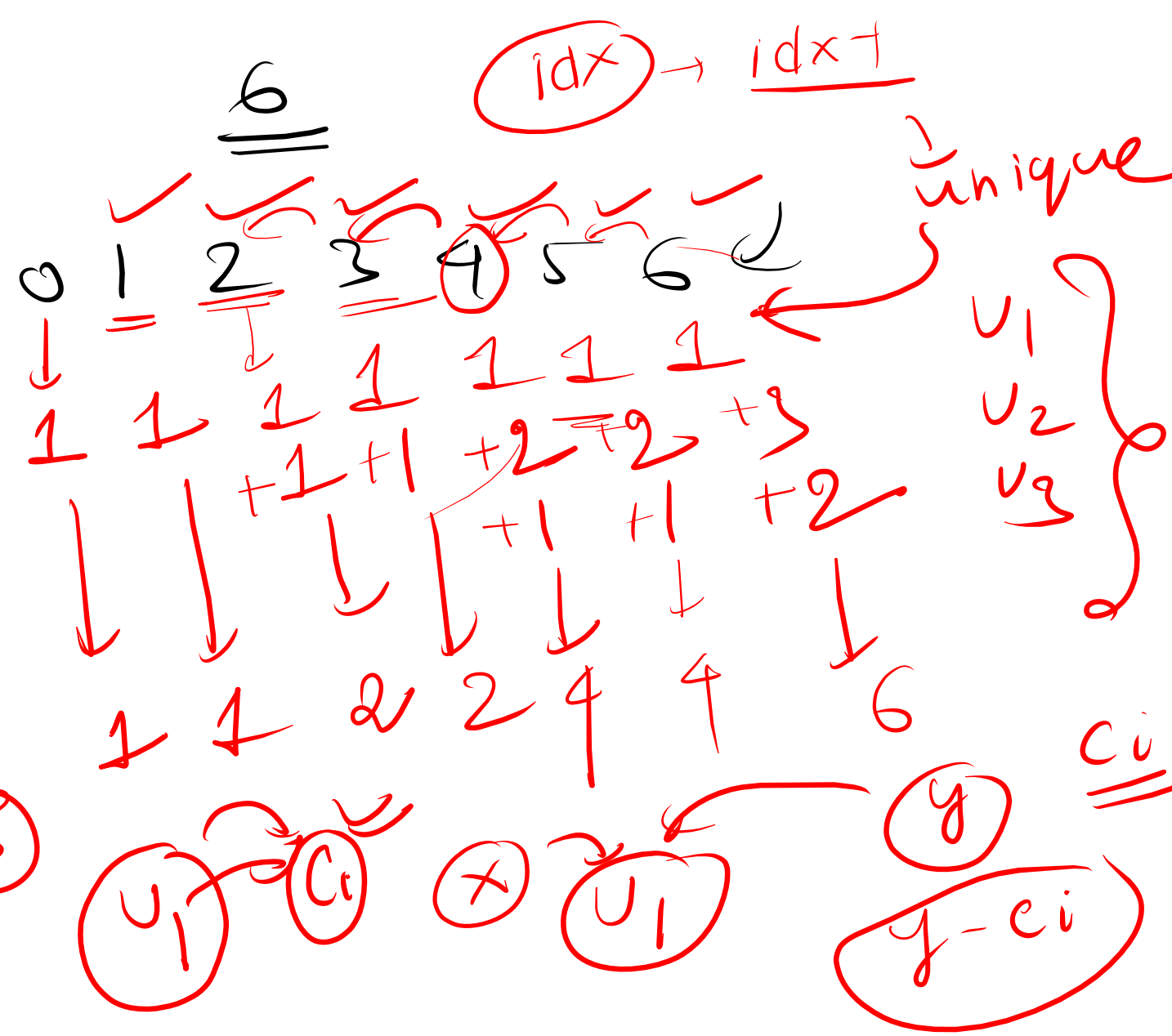
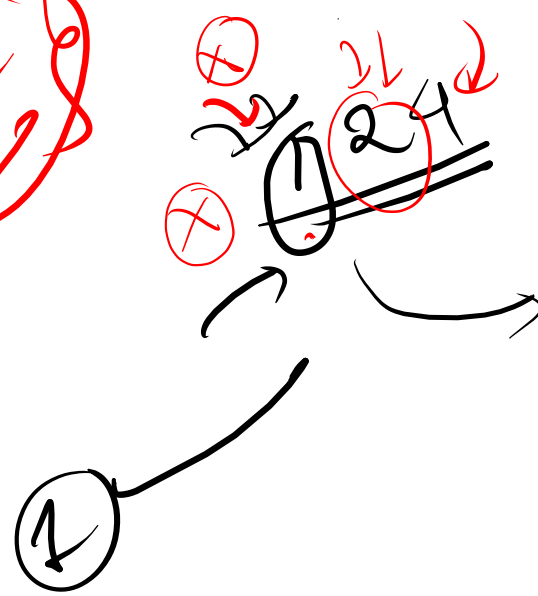
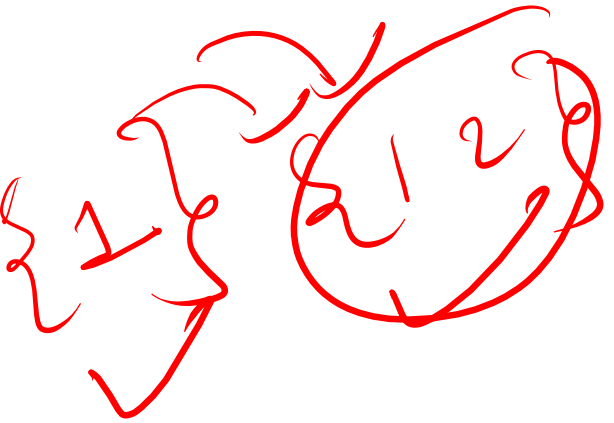
A diagram illustrating a recursive call. A box labeled "sum * n" is shown with arrows pointing to it from above and from the right. A curved arrow points from the box to a square box below it.

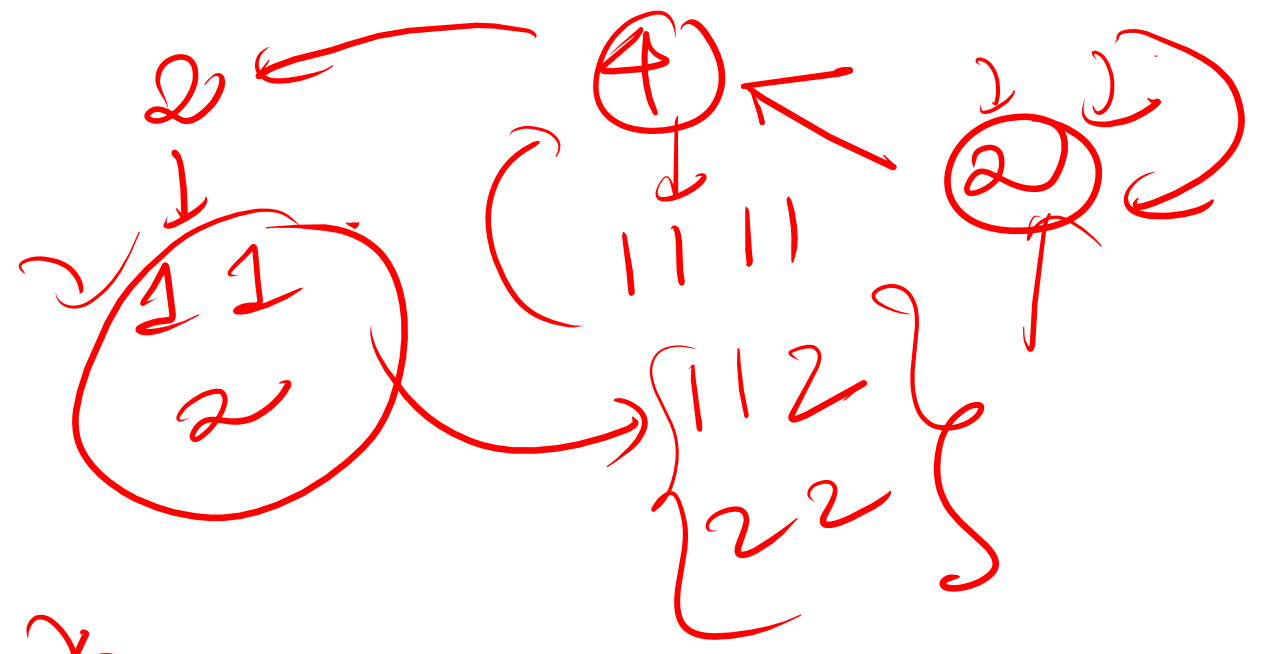
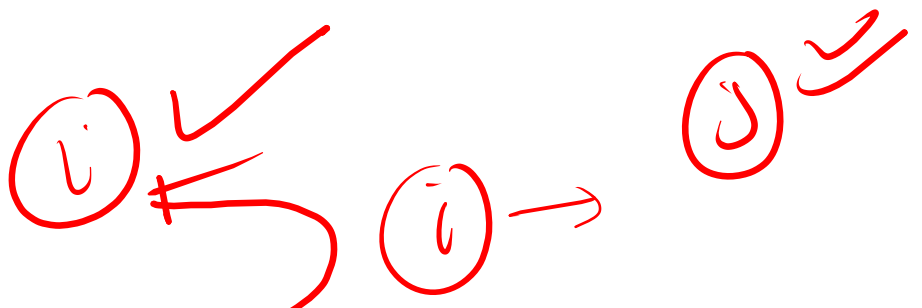


1D
DP

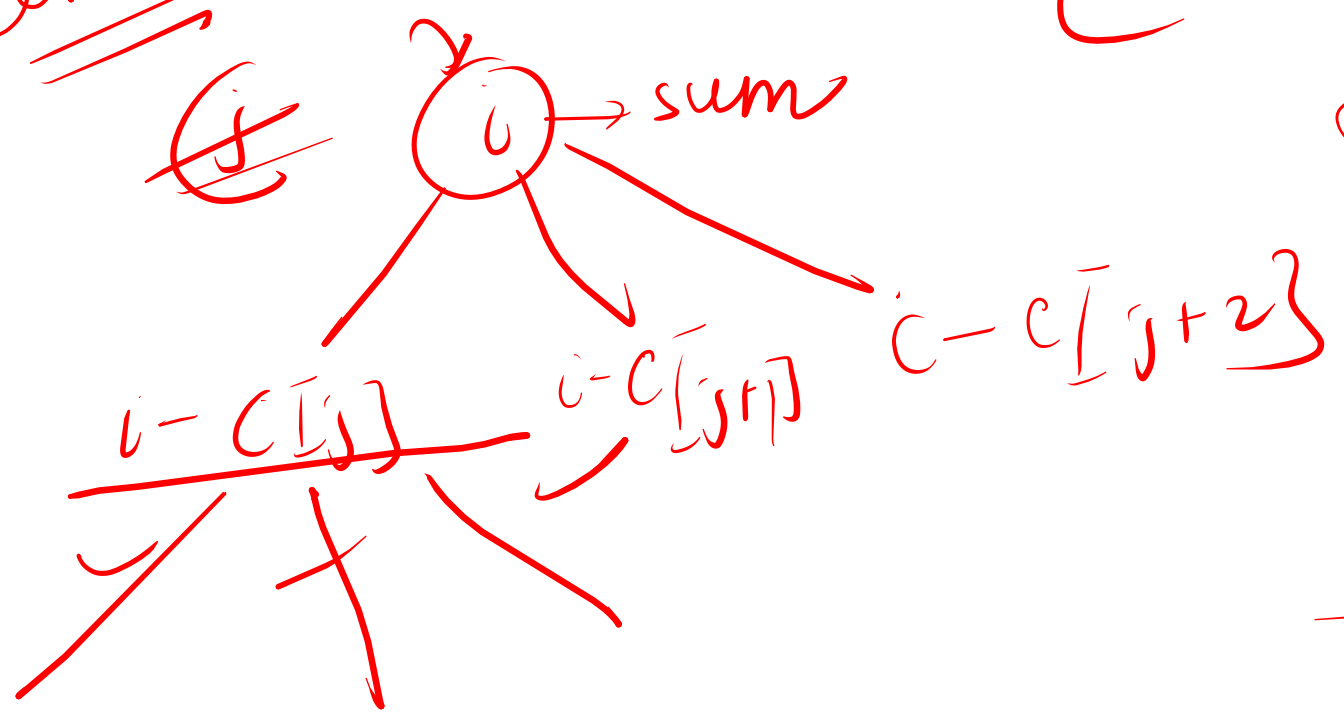
⇒ thought
process







perms



all perms. incl.

1D $\rightarrow O(\text{target})$
sc
 $O(\text{target} * n)$

for($i=0; i(n; i++)$) {

for($j=1; j(\leq \text{target}; j++)$) {

if($c[i] \leq j$)

$dp[i][j] += dp[i - c[i]][j - c[i]]$

}

pick not pick $\rightarrow$ 1 2 3 6 $\Rightarrow$ {1, 2, 3} {6}

sum $\rightarrow$ idx

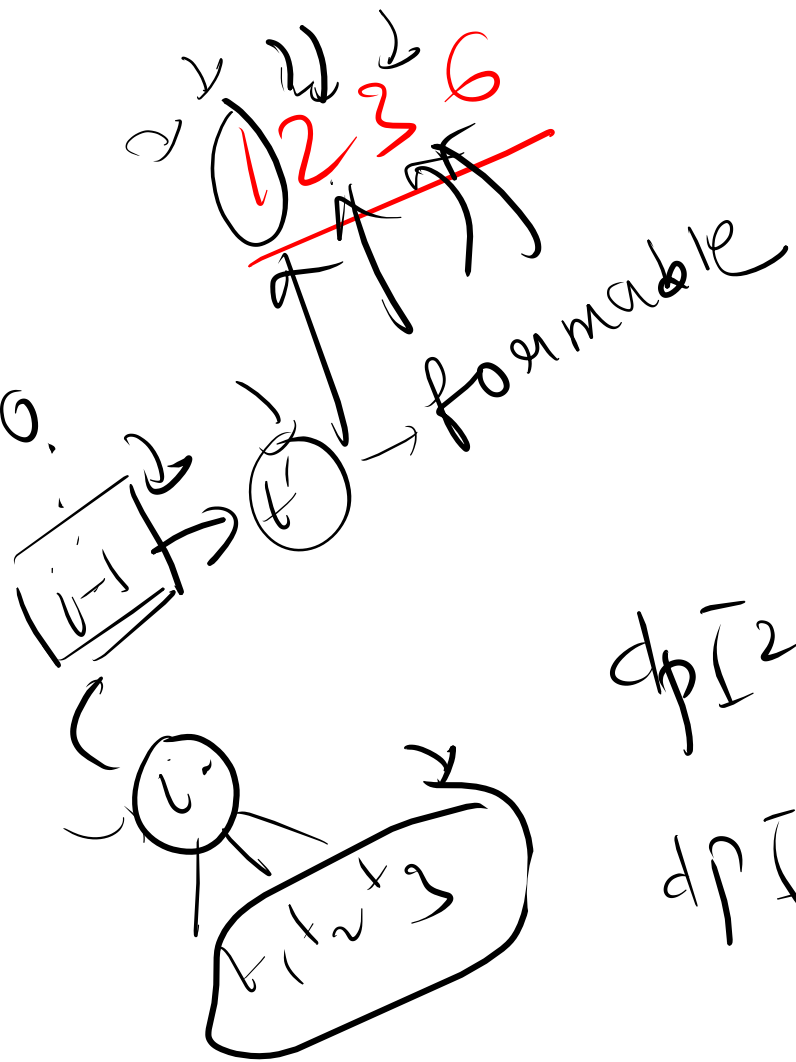
sum
 $\downarrow$
 (sum/2)
 $\downarrow$
 true

0 6

~~12~~

0 1 2 3 4 5 6
 $\downarrow$ $\downarrow$ $\downarrow$
 True True

(2) $\rightarrow$ 2-1 = 1
 $\rightarrow$ {2, 2} {1}



sum $\rightarrow$ idx

sum :

	0	1	2	3	4	5	6
elements idx 0	T	<u>T</u>	F	F	F	F	F
1	T	<u>F</u>	T	T	F	F	F
2	T	T	T	T	T	T	<u>T</u>
3	T						<u>F</u>

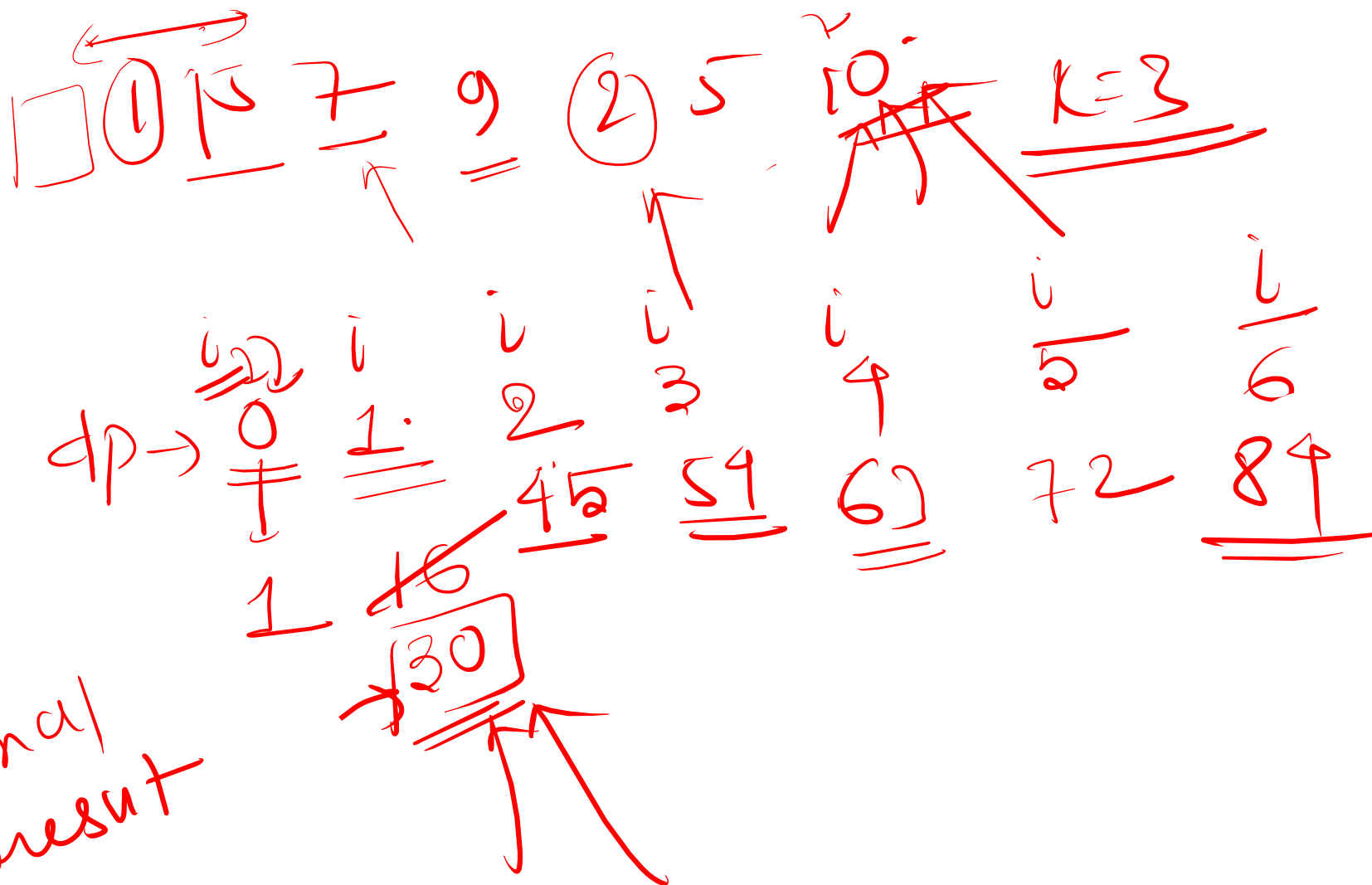
$$dp[2][6] =$$

$$dp[1][3] = \textcircled{1}$$

$\begin{matrix} 0 & 1 \\ 15 & 15 \end{matrix}$

i
 ↘ maximise
 the
 sum

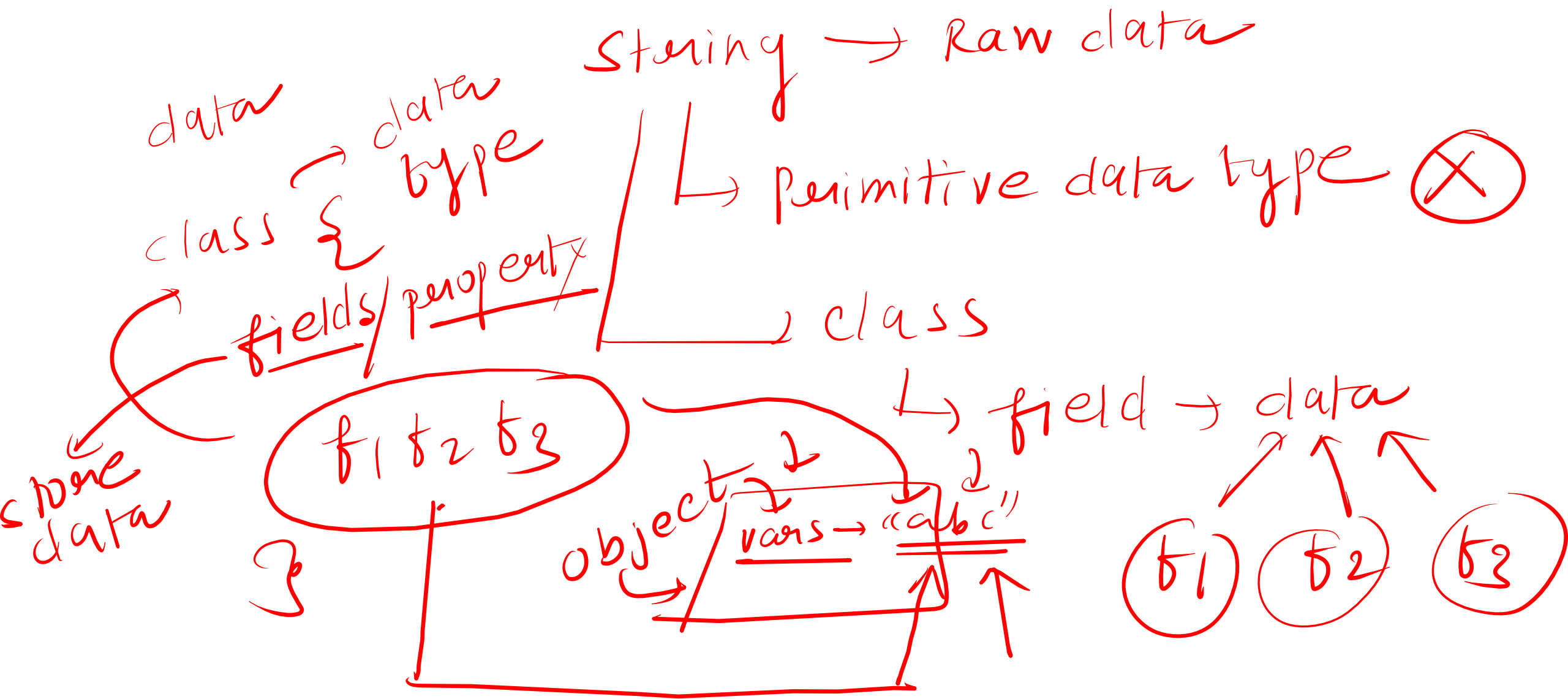
dp[n-1] → final
 result

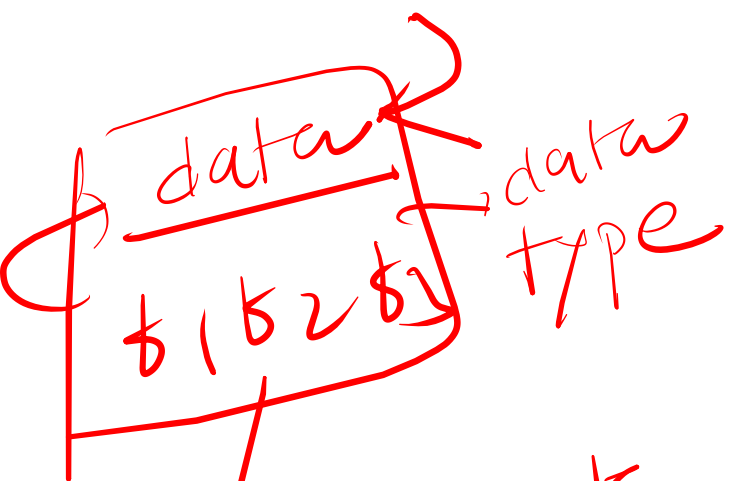


ASCII
↓
"aA126*!"
↑↑

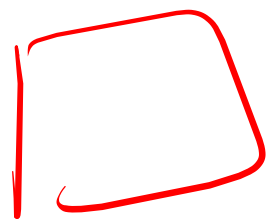
String
↓
"abcde!"
↑↑↑↑↑
→ string

string → Java

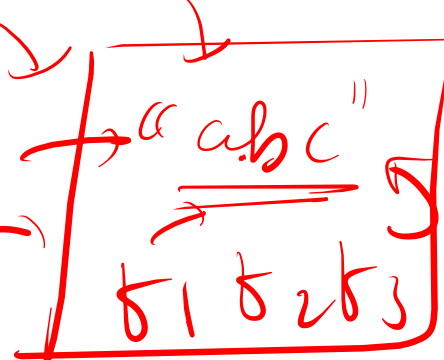




objects



obj →
ref
var

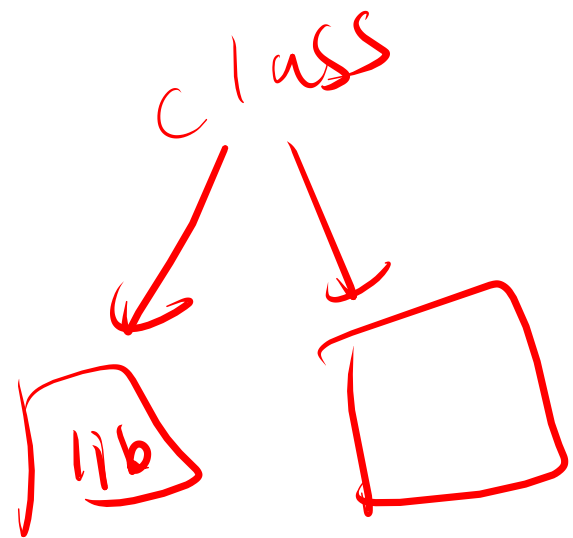


object
of
class
string

var.f1 → abc

var.f2 → 'abc'
count
3

int a = 4
a.add(4)
X



String

↓

class

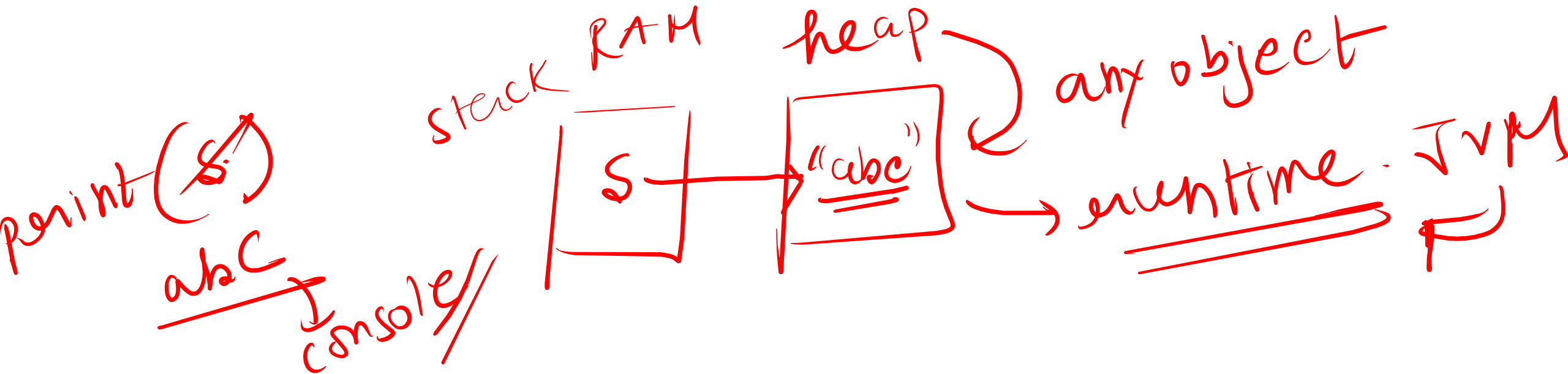
S = "abc"

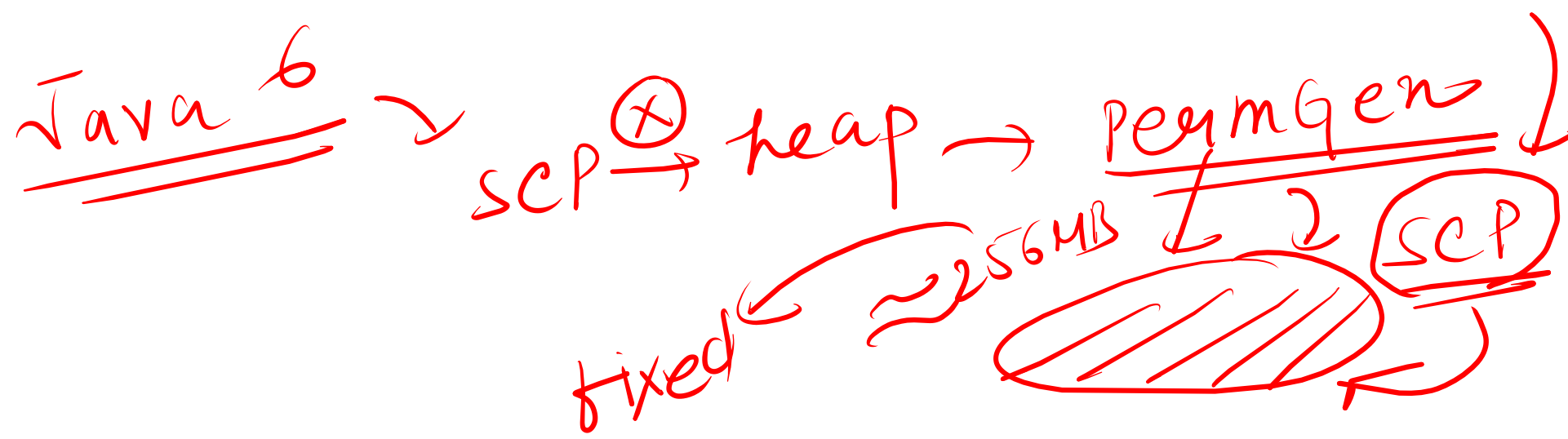
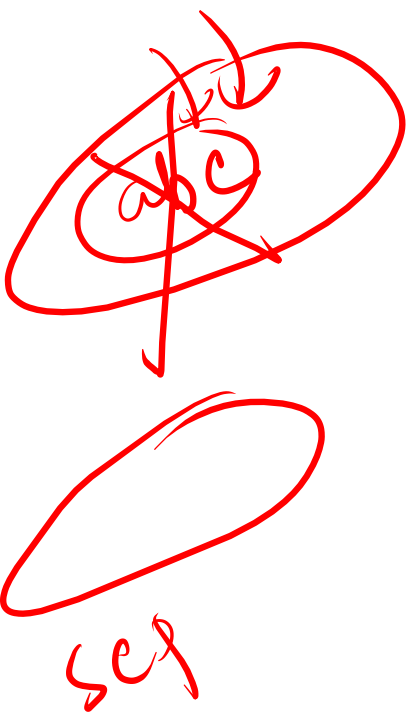
↑

object

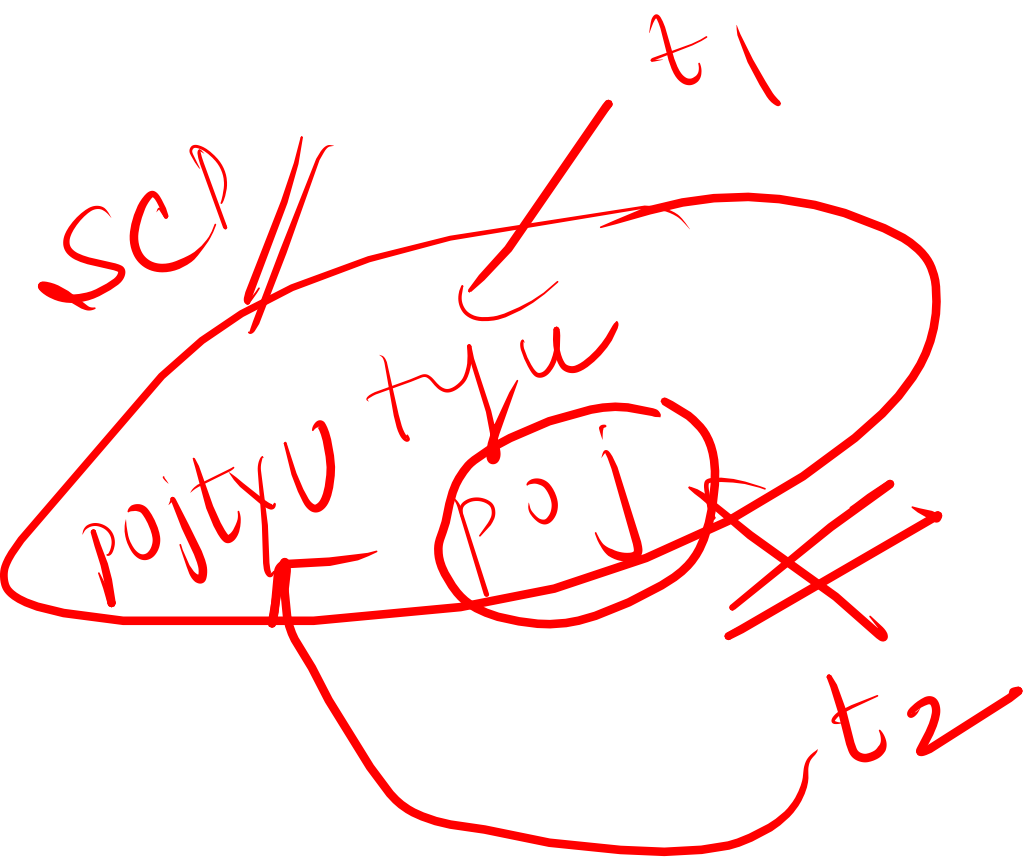
↑

literals

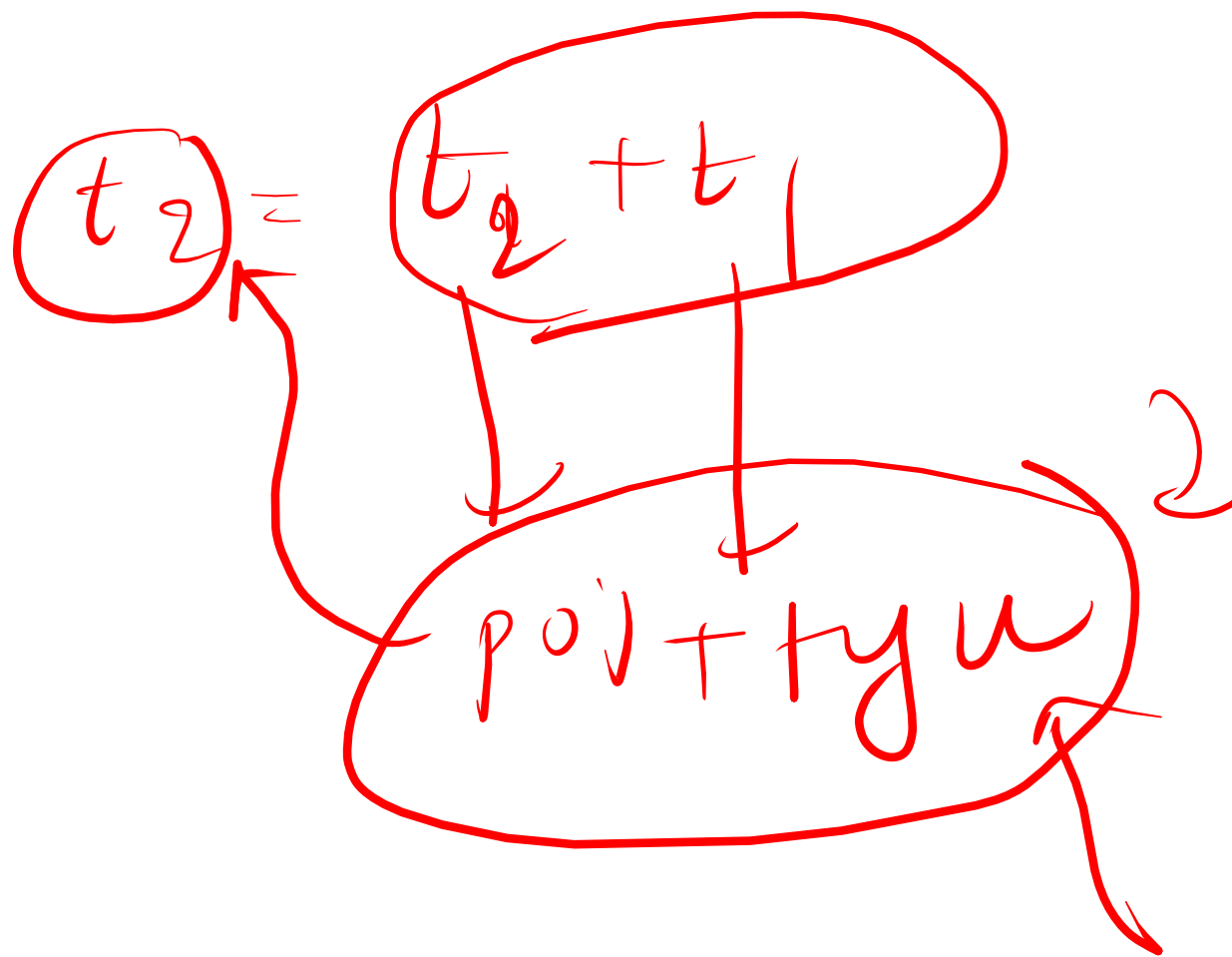




manually develop



$$\left. \begin{array}{l} t_1 = tyu \\ \underline{t_2} = poj \end{array} \right\} sep$$



①
abc

↓
(0,0)

a → (0,1)

ab → (0,2)

abc → (0,3)

b → (1,1)

bc → (1,2)

c → (2,2)

①...①
↓
substring

abcde

↓
substring?

①...①
i i+1 i+2 ... j
↓
substring

n=3
2

$n(n+1)$

$\frac{2}{2}$

$3(4)$

$\frac{2}{2}$
⑥

$$\begin{array}{l} \xrightarrow{n-2} \\ 2 \rightarrow \end{array} \left. \begin{array}{l} c \\ cd \end{array} \right\} \textcircled{2}$$

$$\begin{array}{l} \xrightarrow{n-1} \\ 3 \rightarrow \end{array} \left. \begin{array}{l} d \end{array} \right\} \textcircled{1}$$

$$\begin{array}{l} \xrightarrow{n} \\ \textcircled{n} \end{array} \begin{array}{l} \rightarrow \\ \rightarrow \end{array} \left. \begin{array}{l} n \\ n+1 \\ n+2 \end{array} \right\} \textcircled{2}$$

$$\begin{array}{l} \xrightarrow{n-1} \\ n-1 \rightarrow \end{array} \left. \begin{array}{l} n-2 \end{array} \right\} \textcircled{1}$$

$$\underline{abcd} = \underline{n=4}$$

$$\begin{array}{l} \xrightarrow{0} \\ \xrightarrow{1} \\ \xrightarrow{2} \\ \xrightarrow{3} \end{array} \left. \begin{array}{l} a \\ ab \\ abc \\ abcd \end{array} \right\} \rightarrow \textcircled{n} \textcircled{1}$$

$$\frac{n(n+1)}{2}$$

$$1 \rightarrow \left. \begin{array}{l} b \\ bc \\ bcd \end{array} \right\} \rightarrow \underline{n-1}$$

$$\frac{2}{4} \frac{(5)}{2} \textcircled{10}$$

loop

$i \rightarrow$ ending point.
 $j \rightarrow i$
 $\vdots$

$n-1$

$j \rightarrow \underline{\underline{i/n-1}}$

print
all
substrings

of a
string //

$i=1 \rightarrow b \rightarrow (1,1)$

$i=1 \rightarrow bc \rightarrow (1,2)$

abc

$i=\underline{0} \rightarrow a \rightarrow (0,0)$ $j=0$

$i=0 \rightarrow ab \rightarrow (0,1)$ $j=1$

$i=0 \rightarrow abc \rightarrow (0,2)$ $j=2$

$\swarrow$

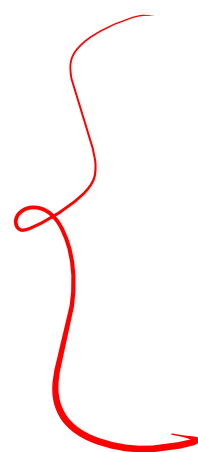
$j \rightarrow (i, n-1)$

s.substring(i, j)

$(i \dots j)$

$$\begin{array}{c} \nearrow \\ i = n-1 \\ \hline \uparrow \end{array}$$

$$\begin{array}{c} \textcircled{+} \\ j \\ \hline \hline n-1 \end{array}$$



$$\begin{array}{c} i \dots j \rightarrow \text{substring}(i, j) \\ \downarrow \qquad \qquad \qquad \searrow \text{excl.} \\ \text{substring}(i, j+1); \qquad i \dots j-1 \end{array}$$

$$\begin{array}{l} \underline{i = 0, \quad j = 0, j = 1 \dots j = n-1} \\ i = 1, \quad j = 1, j = 2 \dots j = n-1 \\ i = 2, \quad j = 2, j = 3 \dots j = n-1 \end{array}$$

$i=1$ $j=3$
 $s = \underline{\underline{a b c d e}}$
 $s.\text{substring}(1, 3)$
 $\swarrow$ $\searrow$
 start end

$\text{for}(i=0; i < n; i++) \{$

$\text{for}(j=i; j < n; j++) \{$

$\text{print}(s.\text{substring}(i, j));$

$\swarrow$
 incl.
 $\text{s.substring}(1, 4)$
 $\searrow$
 excl.

$i \dots j$
 $\downarrow$
 incl.
 $i \dots j$ $(j+1)$

$s = \underline{abcde}$
false $\nwarrow$
 $t = \underline{abd}$

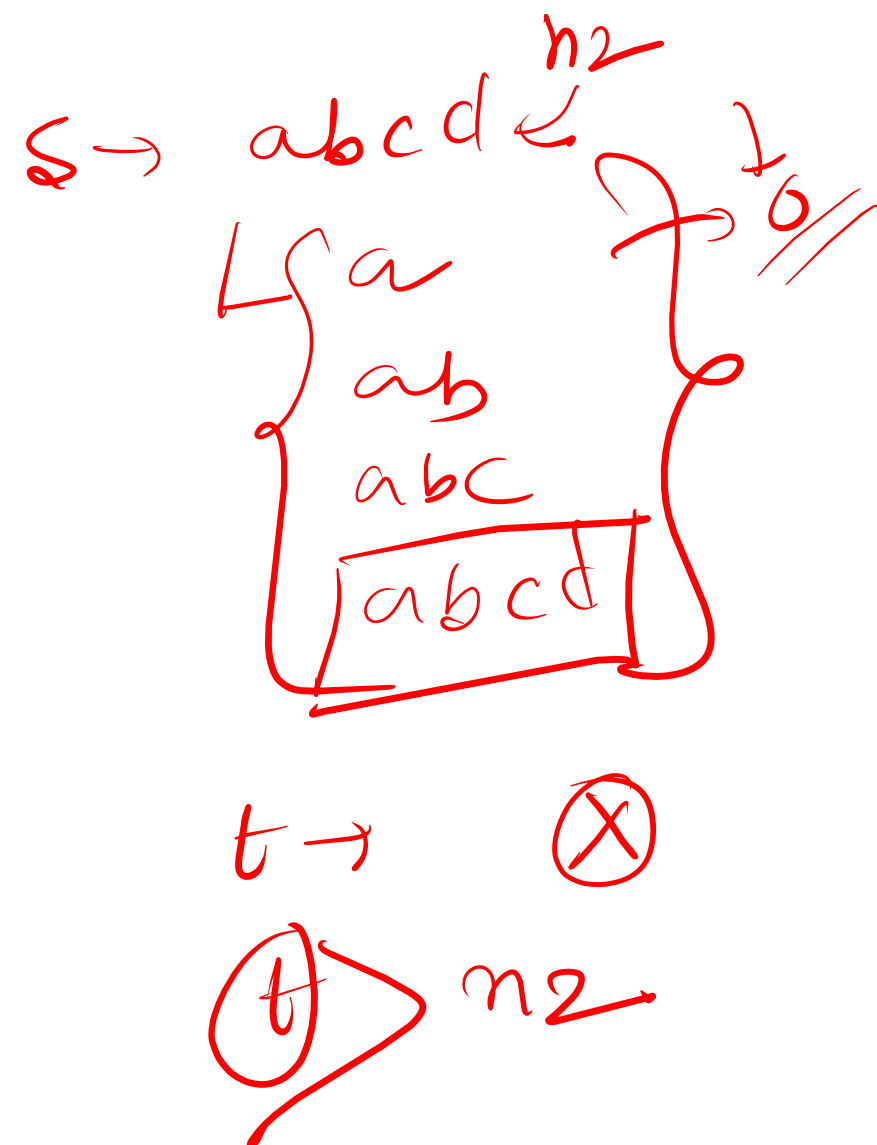
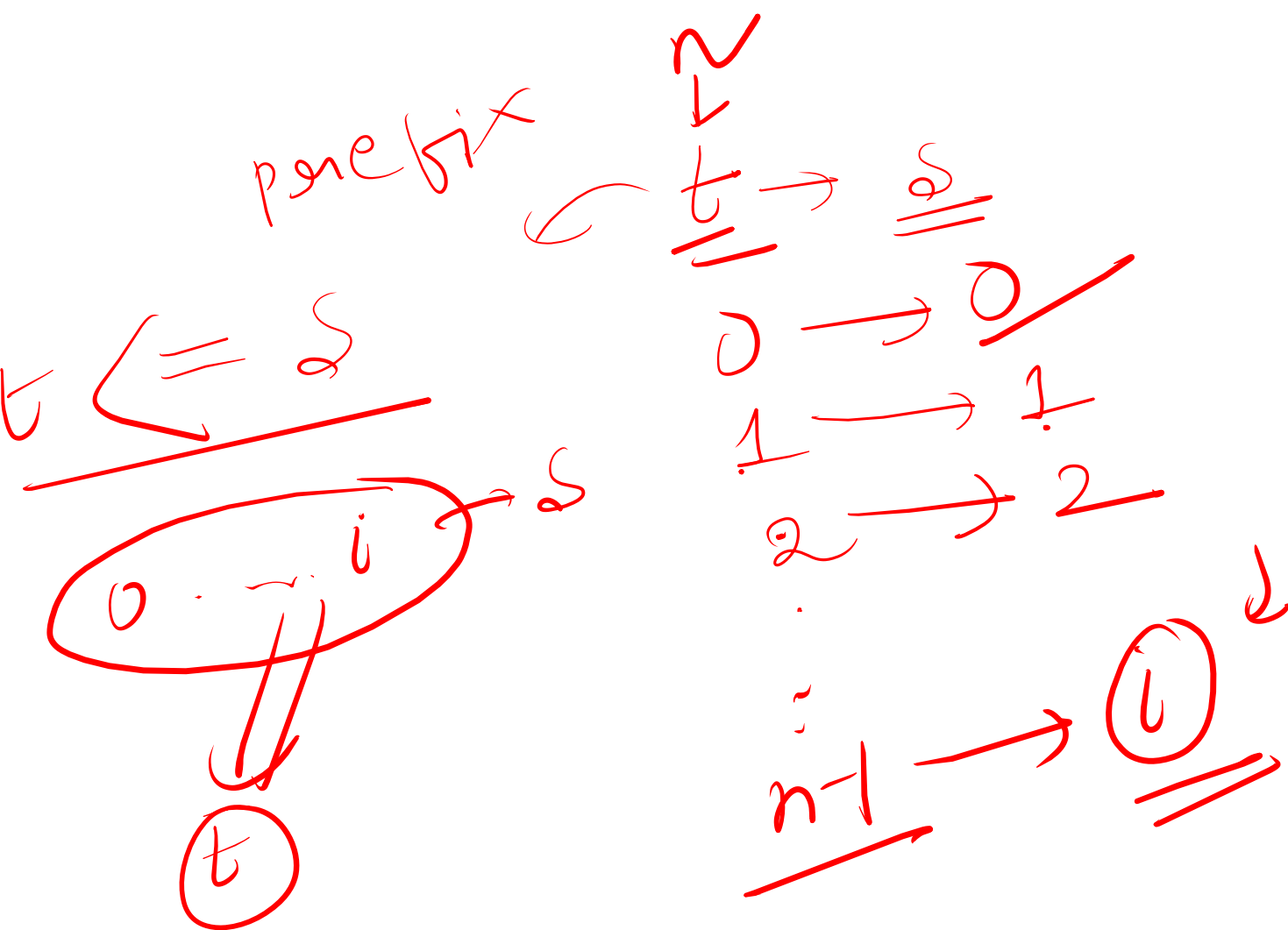
$s = abcde$

$t \rightarrow \boxed{abc}$

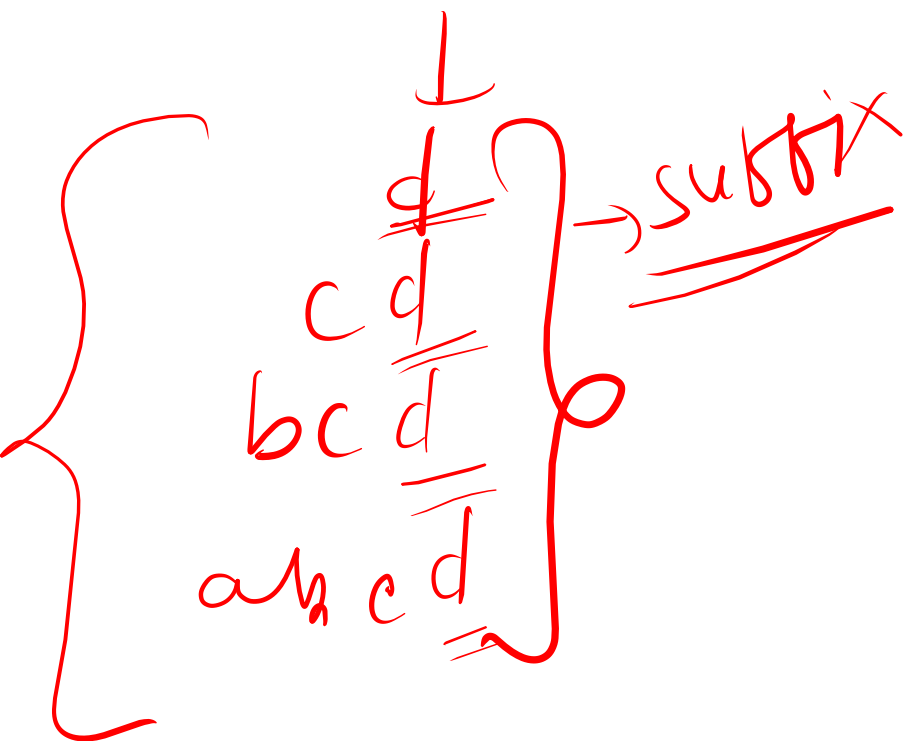
$t \rightarrow$ prefix
 $\downarrow$
 s true.

$s \rightarrow \boxed{abc}de$
 $\downarrow \downarrow \downarrow$

$t \quad a \quad b \quad c \dots nt$
 $\underline{\hspace{10em}}$



s = a b c d



s = a b c d

